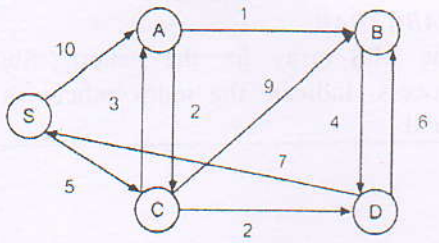
SOMAIYA VIDYAVIHAR UNIVERSITY

02·05·2025(E)

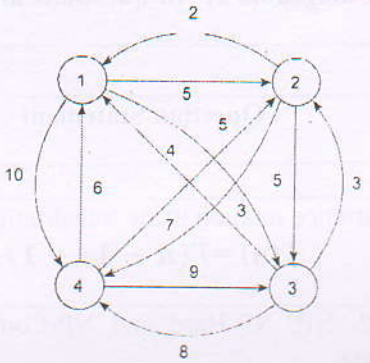
Semester: January 2025– April 2025		
Maximum Marks: 50	**Examination: End-Semester Examination**	**Duration: 2 Hrs.**
Programme code: 01 **Programme:** B.Tech in Computer Engineering	**Class:** SY	**Semester:** IV (SVU-2023)
Institute/School/Department: K. J. Somaiya School of Engineering	**Name of the department/Section/Center:** COMP	
Course Code: 216U01C402	**Name of the Course:** Analysis of Algorithms	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any **two**	
i)	Solve the given recurrence relation using Substitution Method $$T(n) = T(n-1) + 1/n$$	05
ii)	What are classes P, NP, NP-Hard and NP-Complete? Give examples of problems of each class.	05
iii)	Describe Longest common Subsequence problem. Trace the algorithm and find Longest Common Subsequence of following two strings X and Y using dynamic programming. **X = "computer", Y = "commuter"** Compute Length of LCS and also provide all possible LCS strings.	05
Q.2	Attempt any **one**	10
i)	Multiply the following two matrices using Strassen Method: $A = \begin{bmatrix} 3 & 1 \\ 4 & -1 \end{bmatrix}$ $B = \begin{bmatrix} 2 & 6 \\ 6 & -3 \end{bmatrix}$ Find the running time of this operation to perform. Compare Time Complexity of conventional Divide and Conquer method with Strassen method of matrix multiplication.	(6+2+2)
ii)	Apply Dijkstra's single source shortest path for the following graph and find the shortest path between source vertex S and all other vertices.  	

Q.3	Attempt any **one**	**10**
i)	Find the optimal parenthesization of following matrices whose orders are given and perform chain matrix multiplication using the dynamic programming approach.	

A		B		C		D		E	
4 × 5		5 × 3		3 × 2		2 × 7		7 × 2	
p0	p1	p1	p2	p2	p3	p3	p4	p4	p5

ii) Solve the Travelling Salesman Problem using dynamic programming for the following graph. Assume the salesman starts and ends at **city 1**.



Q.4	Attempt the following	**10**
	Explain the Least Cost Branch and Bound (LCBB) approach for solving the 0/1 Knapsack Problem. How does it differ from other methods like Greedy and Dynamic Programming? Solve and draw state space tree generated by LCBB for the following knapsack instance n = 4, (P1, P2, P3, P4) = (10, 10, 12, 18); (w1, w2, w3, w4) = (2, 4, 6, 9) and M = 15.	**(2+2+6)**
Q.5	Attempt the following	**10**
	a. Explain the working of the Knuth-Morris-Pratt (KMP) string matching algorithm. What is the significance of the LPS (Longest Proper Prefix which is also Suffix) array in the algorithm? How does KMP improve over the brute-force pattern matching approach? b. Given the following text and pattern, perform pattern matching using the KMP algorithm: **Text:** *ABABDABACDABABCABAB* **Pattern:** *ABABCABAB* Construct the LPS array for the pattern. Show the step-by-step matching process. Indicate the index/indices in the text where the pattern is found.	**(5+5)**